HOMEWORK 6.7 · UNIT 6 · NC.8.F.3 · NC.8.F.4 · NC.8.F.1

Slope-Intercept Form Day 2

Targets: F.3-4, F.4-4, F.4-5, F.3-1, F.4-6, F.3-7, F.3-6, F.1-4, F.4-8, F.3-9

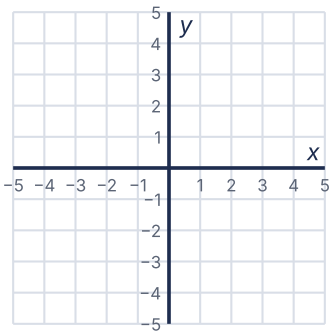
Name _____ Date _____ Block _____

Remember: In $y = mx + b$, m = slope = $\frac{\text{rise}}{\text{run}}$ and b = y-intercept (where the line crosses the y-axis). **To graph** $y = 2x - 3$: plot (0, −3), then go up 2 and right 1 to (1, −1). In a story, m = rate of change (per hour, per week…) and b = the starting amount.

SCORE 3.0 Approaching · the basics

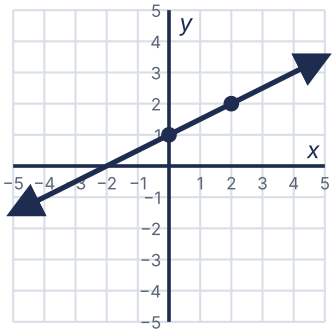
Targets: F.3-4, F.4-4, F.4-5, F.3-1

1 Graph $y = -x + 2$ on the grid. F.3-4



Hint: Start at b on the y-axis.

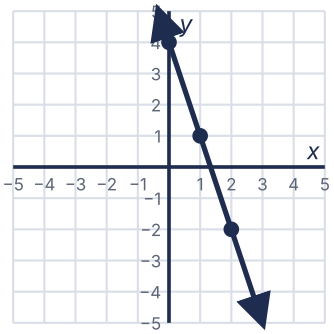
2 Which equation matches the graph? F.3-4



Ⓐ $y = \frac{1}{2}x + 1$ Ⓑ $y = 2x + 1$

Ⓒ $y = x + \frac{1}{2}$ Ⓓ $y = -\frac{1}{2}x + 1$

3 Write the equation of the line in slope-intercept form. F.4-4



_____

4 Find the slope of the line through (−2, 5) and (2, −3). REVIEW 6.5 F.4-5

Ⓐ −2 Ⓑ 2 Ⓒ $-\frac{1}{2}$ Ⓓ −8

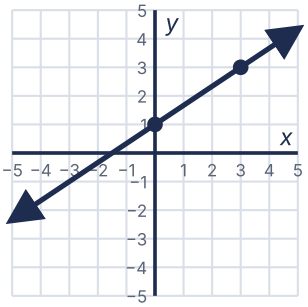
5 The graph of $y = x^2 + 3$ is a straight line. REVIEW 6.4 F.3-1

True False

SCORE 4.0 Proficient · on target Targets: F.4-6, F.3-7, F.3-6, F.1-4

6 A candle is 12 inches tall. It burns down 2 inches each hour. Write an equation for its height y after x hours. What does the slope mean? F.4-6,F.3-7

7 **Mei** wrote the equation of this line. Find her mistake. Fix it. F.3-6



MEI'S WORK

rise = 3, run = 2

$y = \frac{3}{2}x + 1$

8 Give the domain and the range: $\{(1, 4), (2, 6), (3, 4)\}$ REVIEW 6.1 F.1-4

SCORE 5.0 Mastery · go deeper Targets: F.4-8, F.3-9

9 A gym charges a sign-up fee plus a monthly fee. 3 months cost $110. 7 months cost $190. Write an equation. What does ***b*** mean? F.4-8

The rate is _____ per month. b = _____ , which means _____ .

10 This table is linear. Write its equation. Explain each step. F.3-9

x	1	3	5
y	1	4	7

The slope is _____ because _____ . The y-intercept is _____ because _____ .